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DA5000W Linear Algebra & Optimization
Assignment 1 – October 2025

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Q1 Find solution for linear system of equations:

$$-2x + 3y + z = 1$$
$$-x + z = 0$$
$$2y = 5$$

Ans 1. $2y = 5 \Rightarrow y = \frac{5}{2}$

$\therefore -2x + 3\left(\frac{5}{2}\right) + z = 1$

$\Rightarrow -2x + z = -\frac{13}{2}$

But, $-x + z = 0 \Rightarrow x = z$

$\therefore -2x + x = -\frac{13}{2} \Rightarrow x = z = \frac{13}{2}$

Q2 Find basis of null space and its dimensions:

$$x + 5y + 2z$$
$$2x + 4y + 3z$$
$$4x + 8y + 6z$$

Ans Coefficient matrix:

$$A = \begin{bmatrix} 1 & 5 & 2 \\ 2 & 4 & 3 \\ 4 & 8 & 6 \end{bmatrix}$$

Row Reduce: $R_3 \to R_3 - 2R_2$

$$A = \begin{bmatrix} 1 & 5 & 2 \\ 2 & 4 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\cancel{\therefore A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0 \Rightarrow \begin{matrix} x + 5y + 2z = 0 \\ 2x + 4y + 3z = 0 \end{matrix}}$$

Row Reduce: $R_2 \to R_2 - 2R_1$

$$A = \begin{bmatrix} 1 & 5 & 2 \\ 0 & -6 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore x + 5y + z = 0$$
$$-6y - z = 0$$

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Let free variable $z = t$.

$\therefore -6y - t = 0 \Rightarrow y = \frac{-t}{6}$

$x + 5\left(\frac{-t}{6}\right) + 2t = 0 \Rightarrow x = \frac{-7t}{6}$

So $(x, y, z) = \left(\frac{-7t}{6}, \frac{-t}{6}, t\right)$

$\therefore$ Set of basis vectors is.

$$\left\{ \begin{bmatrix} -7/6 \\ -1/6 \\ t \end{bmatrix} \right\} \text{ (only 1)}$$

So dimension of null space = 1

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Q3. Decide whether or not following vectors are linearly independent:

(a) $\vec{v_1} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ $\vec{V_2} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$

$\vec{V_3} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ $\vec{V_4} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

Ans Form a matrix using vectors as
(a) columns:

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\det(A) = 1[0+0+1(0\times 1-1)] - 1[1(1\times 1 - 1\times 0)+0+1(0-0)] + 0 + 0$$

$$= -1+1 = 0$$

Since, $\det(A) = 0$,

$\therefore$ Vectors are linearly dependent

(b) Do vectors span R^4, by trying to solve $C_1\vec{V_1} + C_2\vec{V_2} + C_3\vec{V_3} + C_4\vec{V_4} = (0, 0, 0, 1)$

Ans (b) writing in matrix form:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow C_1 + C_2 = 0 \Rightarrow C_2 = -C_1$$
$$C_1 + C_4 = 0 \quad C_4 = -C_1$$
$$C_2 + C_3 = 0 \quad C_3 = -C_2 = C_1$$
$$C_3 + C_4 = 0 \quad C_4 = -C_3 = -C_1$$

$\therefore$ C_1 is free variable, hence vectors do not span R^4

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Q4. Let A be a 3×7 matrix, with rank 3. Comment on nature of solutions of A.

Ans rank(A) = no. of rows in A = 3, so equations are consistent.

No. of free variables = $7-3=4$

Since free vars are more than 0, ∴ infinite solutions exist.

Q5 Let $T: R^4 \to R^3$ ~~be~~ matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \end{bmatrix}$$

Calculate rank, nullity and find a basis for the null space.

Ans $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \end{bmatrix}$

$R_3 \rightarrow R_3 - R_1 + R_2$

$\Rightarrow A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$\therefore$ rank$(A) = 3$

To find basis of null space:

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

~~Let C_3, C_4 be free variables.~~

$C_1 + 2C_2 + 3C_3 + 4C_4 = 0$

$\Rightarrow C_1 + 2(-C_3) + 3C_3 + 4(0) = 0$

$\Rightarrow C_1 = -C_3$

$C_2 + C_3 + 2C_4 = 0 \Rightarrow C_2 + C_3 + 2(0) = 0$

$C_4 = 0$

$\Rightarrow C_2 = -C_3$

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So in terms of free variables C_3, vector becomes:

$$\begin{bmatrix} -C_3 \\ -C_3 \\ C_3 \\ 0 \end{bmatrix} \longrightarrow \underline{\text{Basis vector (only 1)}}$$

So, nullity = no. of basis vectors

$\Rightarrow$ <u>nullity = 0</u>

Q6 Is each statement true or false?

(1) vectors $\vec{b}$ that are not in the column space $C(A)$ form a subspace.

(2) If $C(A)$ contains only zero vector, A is square matrix.

(3) Column space of $2A$ equals column space of A

(4) Column space of $A - I$ equals column space of A.

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Ans (a) Column space of matrix is:

$$C(A) = \{Ax : x \in R^n\}$$

Subspace conditions are:

→ must contain zero vector

→ must be closed under vector addition

→ must be closed under scalar multiplication

0 vector is in C(A) because

$A \cdot 0 = 0$

So 0 vector is not present in ~~subspac~~ set of column vectors not in C(A), so it does not form a ~~column~~ subspace. $\Rightarrow$ $\boxed{\text{FALSE}}$

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(2) If column space $C(A)$ only contains ~~column~~ zero vector, then all columns of A must be 0. But it can be rectangular, not necessarily square. So statement is $\boxed{\text{FALSE}}$

(3) $C(A) = \{Ax : x \in R^n\}$

$C(2A) = \{2Ax : x \in R^n\}$

Let $y = 2x$

$\Rightarrow C(2A) = \{Ay : y \in R^n\}$

But this is same as $C(A)$.

$\therefore C(2A) = C(A) \rightarrow \boxed{\text{TRUE}}$

(4) $C(A-I) = C(A) - C(I)$

But column space of I,

$C(I) = R^n$ (all vectors)

$\therefore C(A-I) = C(A)$ [which is subset of R^n]

$\boxed{\text{TRUE}}$

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Q7 In linear system of equations

$$x + 2y - z = 3$$
$$2x - y + 3z = 1$$
$$4x + 3y + z = k$$

(1) Find values of constant k for which system is consistent (at least 1 solution)

(2) For value of k in (1), what is rank of coefficient matrix and rank of augmented matrix? Comment on nature of solution (unique or infinite solutions)

Ans (1)

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -1 & 3 & 1 \\ 4 & 3 & 1 & k \end{array}\right]$$

(Augmented matrix)

Row Reduce: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 4R_1$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -5 & 5 & -5 \\ 0 & -5 & 5 & k-12 \end{array}\right]$$

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Now coefficients of R_2, R_3 are same, so for consistent system,

$k - 12 = -5 \Rightarrow \boxed{k = 7}$

(2) Since system is consistent,
rank (coeff matrix) = rank (augmented)
$= \underline{2}$ (no. of independent rows)

Infinite no. of solutions exist, since rank (2) < no. of variables (3).

Q8 Let $\vec{v} = \begin{bmatrix} 3 \\ -1 \\ 2 \end{bmatrix} \in R^3$

and let $B = \{\vec{b_1}, \vec{b_2}, \vec{b_3}\}$ be an orthonormal basis of R^3 defined by:

$b_1 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$ $b_2 = \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$ $b_3 = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{bmatrix}$

Find coordinate vector $[\vec{v}]_B$ of $\vec{v}$ with respect to orthonormal basis B.

Ans $\vec{v_1}\cdot\vec{b_1} = 3\left(\frac{1}{\sqrt{3}}\right) - 1\left(\frac{1}{\sqrt{3}}\right) + 2\left(\frac{1}{\sqrt{3}}\right) = \frac{4}{\sqrt{3}}$

$\vec{v}\cdot\vec{b_2} = 3\left(\frac{-1}{\sqrt{2}}\right) - 1\left(\frac{1}{\sqrt{2}}\right) + 2(0) = -2\sqrt{\;}$

$\vec{v}\cdot\vec{b_3} = 3\left(\frac{1}{\sqrt{6}}\right) - 1\left(\frac{1}{\sqrt{6}}\right) + 2\left(\frac{-2}{\sqrt{6}}\right) = -\frac{\;}{\sqrt{\;}}$

$\therefore$ Coordinate vector wrt basis B is:

$$[\vec{v}]_B = \begin{bmatrix} 4/\sqrt{3} \\ -2\sqrt{2} \\ -2/\sqrt{6} \end{bmatrix}$$

Q9 Let $T: R^3 \to R^3$ be:

$T(x, y, z) = (2x - y - z,\ 3x + y - 2z,\ x + 4y + z)$

(1) matrix representation of T in standard basis

(2) image of standard basis vectors under T

(3) verify T satisfies linear transform properties

(4) kernel, image of T? verify rank - nullity relation

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(5) Is T invertible? Comment geometrically.

Ans (1) $\begin{bmatrix} 2 & -1 & -1 \\ 3 & 1 & -2 \\ 1 & 4 & 1 \end{bmatrix}$ is matrix repr of T

(2) Images of standard basis vectors:

$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ $Te_1 = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$

$e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ $Te_2 = \begin{bmatrix} -1 \\ 1 \\ 4 \end{bmatrix}$

$e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ $Te_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$

(3) Properties of linear transform:

(a) additive :

$T(\vec{U}) + T(\vec{V})$

$= \begin{bmatrix} 2U_x - \cancel{2}U_y - U_z \\ 3U_x + U_y - 2U_z \\ U_x + 4U_y + U_z \end{bmatrix} + \begin{bmatrix} 2V_x - V_y - V_z \\ 3V_x + V_y - 2V_z \\ V_x + 4V_y + V_z \end{bmatrix}$

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$$= \begin{bmatrix} 2(U_x+V_x)-(U_y+V_y)-(U_z+V_z) \\ 3(U_x+V_x)+(U_y+V_y)-2(U_z+V_z) \\ (U_x+V_x)+4(U_y+V_y)+(U_z+V_z) \end{bmatrix}$$

$= T(\vec{U}+\vec{V})$

$\therefore$ additive property $T(\vec{U}) + T(\vec{V}) = T(\vec{U}+\vec{V})$

is satisfied.

(b) homogenous: (multiply with scalar):

$$cT(\vec{v}) = \begin{bmatrix} 2cV_x - cV_y - cV_z \\ 3cV_x + cV_y - 2cV_z \\ cV_x + 4cV_y + cV_z \end{bmatrix}$$

$= T(c\vec{V})$

$\therefore$ homogenous property is satisfied.

(c) kernel $= \{\vec{V} \mid A\vec{v} = \vec{0}\}$

so $\begin{bmatrix} 2 & -1 & -1 \\ 3 & 1 & -2 \\ 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

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$R_1 \leftrightarrow R_3$ swap

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 0 \\ 3 & 1 & -2 & 0 \\ 2 & -1 & -1 & 0 \end{array}\right]$$

$R_2 \to R_2 - 3R_1$, $R_3 \to R_3 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 0 \\ 0 & -11 & -5 & 0 \\ 0 & -3 & 5 & 0 \end{array}\right]$$

$R_2 \to -1/11\, R_2$

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 0 \\ 0 & 1 & 5/11 & 0 \\ 0 & -3 & 5 & 0 \end{array}\right]$$

$R_3 \to R_3 + 3R_2$

$$\left[\begin{array}{ccc|c} 1 & 4 & 1 & 0 \\ 0 & 1 & 5/11 & 0 \\ 0 & 0 & 70/11 & 0 \end{array}\right]$$

$\therefore \quad x + 4y + z = 0 \quad \Rightarrow x = y = z = 0$

$y + \frac{5}{11} z = 0$

$\frac{70}{11} z = 0$

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So kernel has only zero vector: $\{\vec{0}\}$

Image of T = $\begin{bmatrix} 2x - y - z \\ 3x + y - 2z \\ x + 4y + z \end{bmatrix}$ forall $x, y, z \in R^3$

Nullity = no. of vectors in kernel/null space ~~Rank~~ except zero vector
= 0

Rank = dimension of image space = 3

No. of rows in T (3) = rank (3) + nullity (0)

∴ Rank - Nullity property verified.

(5) T is full-rank, so it is invertible. Geometrically it means that we can map result vectors back to input vectors using inverse T^{-1}.

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Q10 Let W be subspace of R^3 spanned by orthonormal vector

$$\vec{U_1} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \quad \vec{U_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

(1) Find orthogonal projection of vector $\vec{v} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$ onto subspace.

(2) Find distance between vector $\vec{v}$ and subspace W.

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(1) Orthogonal Projection (closest vector present in subspace) is:

$$\vec{v_w} = \frac{\vec{v}\cdot\vec{U_1}}{\cancel{\|\vec{U_1}\|^2}}\vec{U_1} + \frac{\vec{v}\cdot\vec{U_2}}{\cancel{\|\vec{U_2}\|^2}}\vec{U_2}$$

(since $\vec{U_1}, \vec{U_2}$ are orthonormal/unit vectors)

$$\vec{v_1}\cdot\vec{U_1} = 3\left(\frac{1}{\sqrt{2}}\right) + 2\left(\frac{1}{\sqrt{2}}\right) + 5(0) = \frac{5}{\sqrt{2}}$$

$$\vec{v_1}\cdot\vec{U_2} = 3(0) + 2(0) + 5(1) = 5$$

$$\vec{v_w} = \frac{5}{\sqrt{2}}\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + 5\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5/2 \\ 5/2 \\ 5 \end{bmatrix}$$

(2) distance between $\vec{v}$ and W

$$= |\vec{v} - \vec{v_w}| = \sqrt{(3-5/2)^2 + (2-5/2)^2 + (5-5)^2}$$

$$= \frac{1}{\sqrt{2}}$$